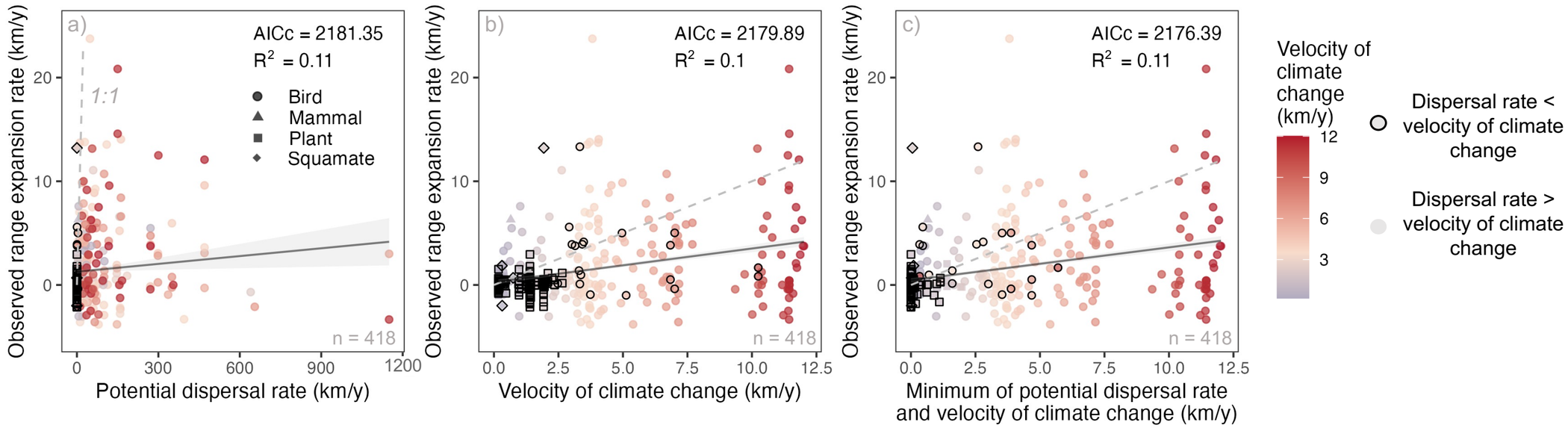
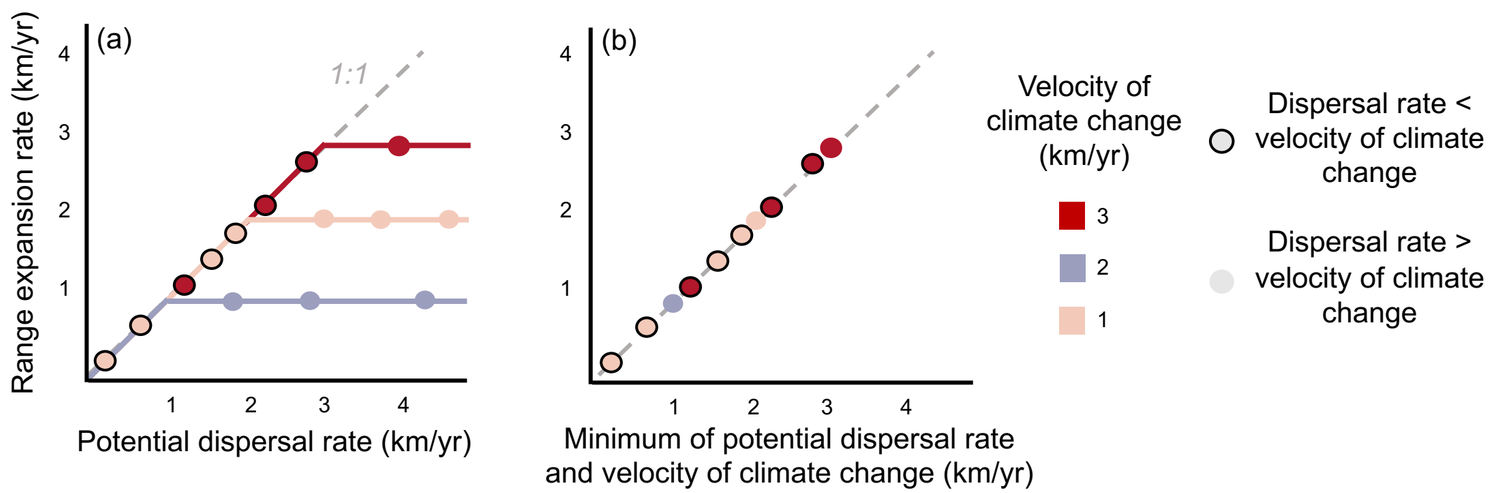
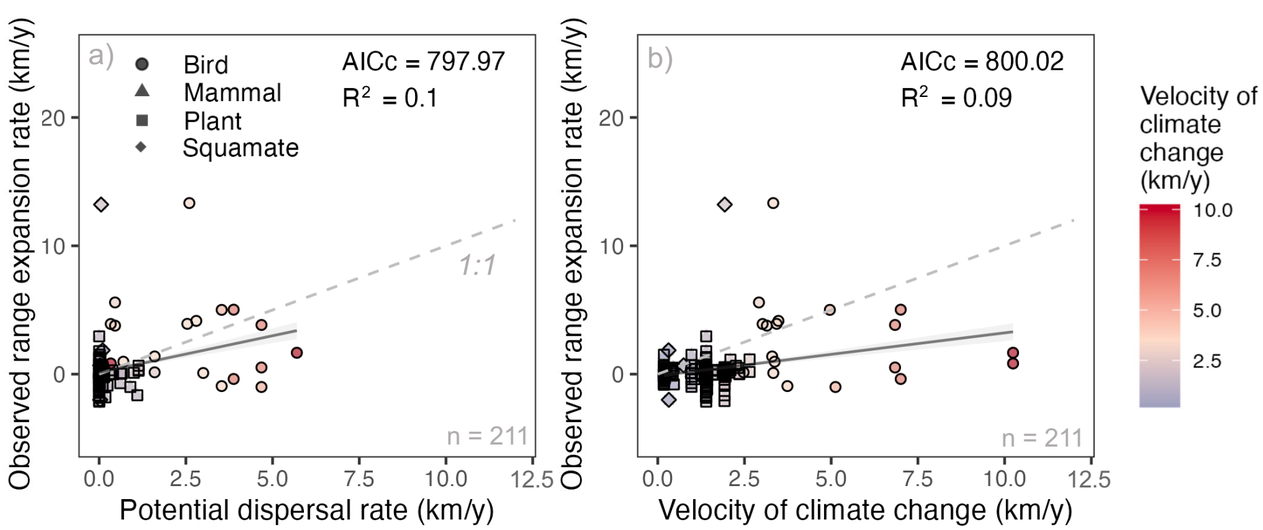
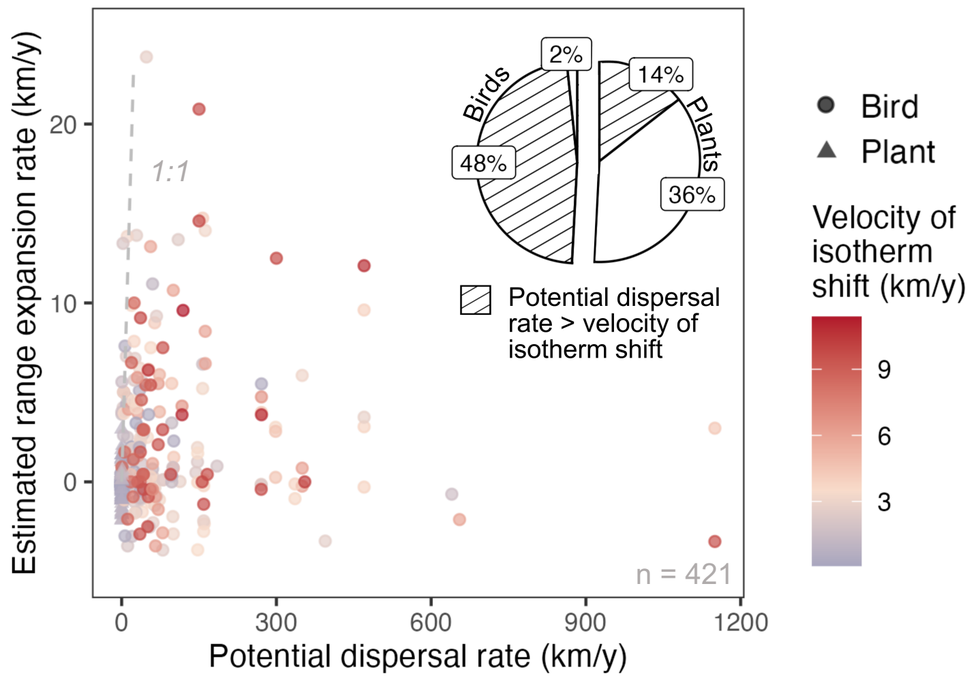
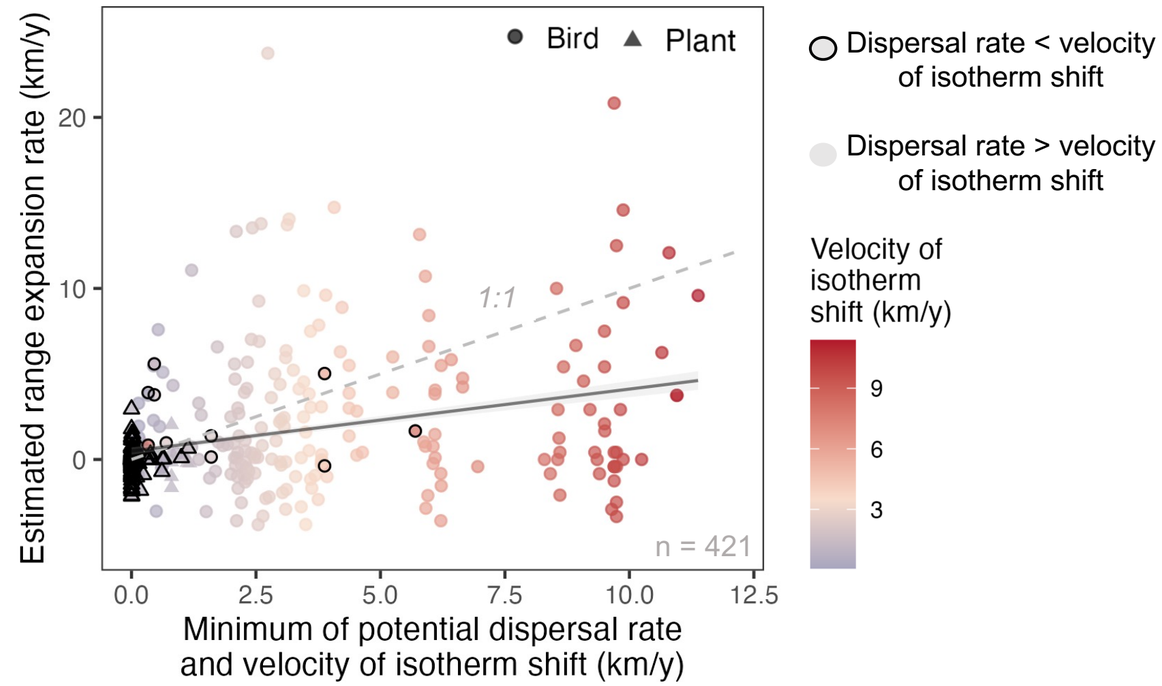
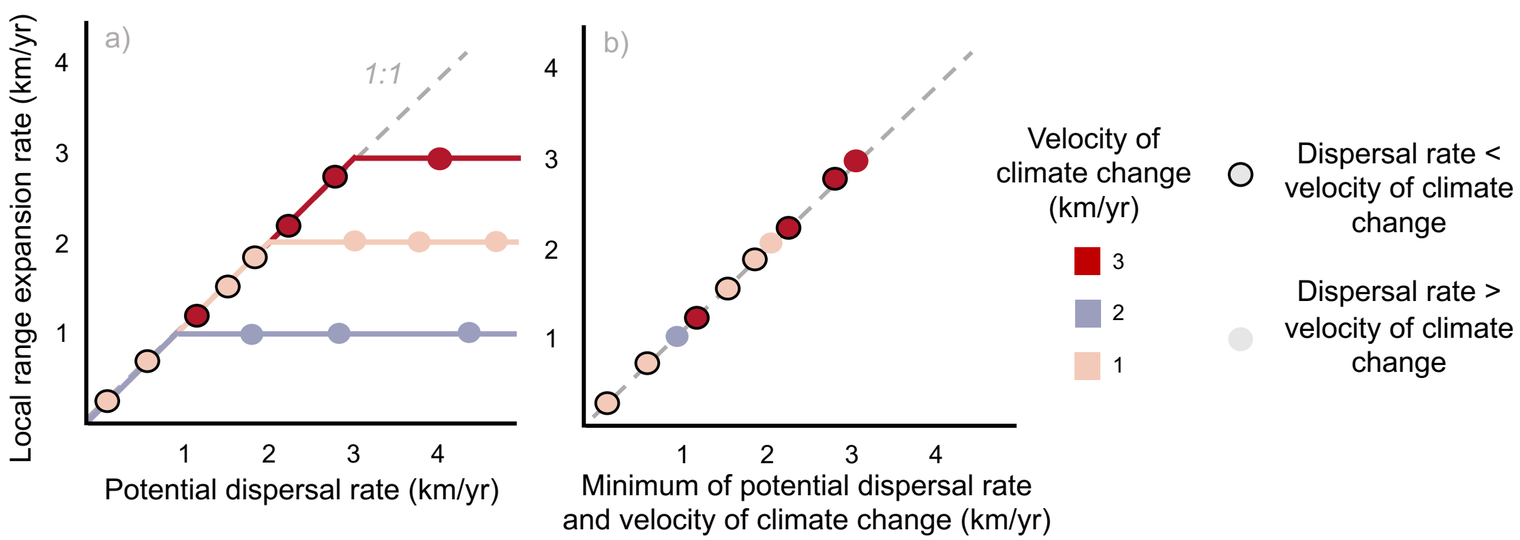

a)
AICc = 2181.35
R^2 = 0.11
Bird
Mammal
Plant
Squamate
1:1
n = 418
Observed range expansion rate (km/y)
Potential dispersal rate (km/y)
b)
AICc = 2179.89
R^2 = 0.1
n = 418
Velocity of climate change (km/y)
c)
AICc = 2176.39
R^2 = 0.11
n = 418
Minimum of potential dispersal rate and velocity of climate change (km/y)
Velocity of climate change (km/y)
Dispersal rate < velocity of climate change
Dispersal rate > velocity of climate change

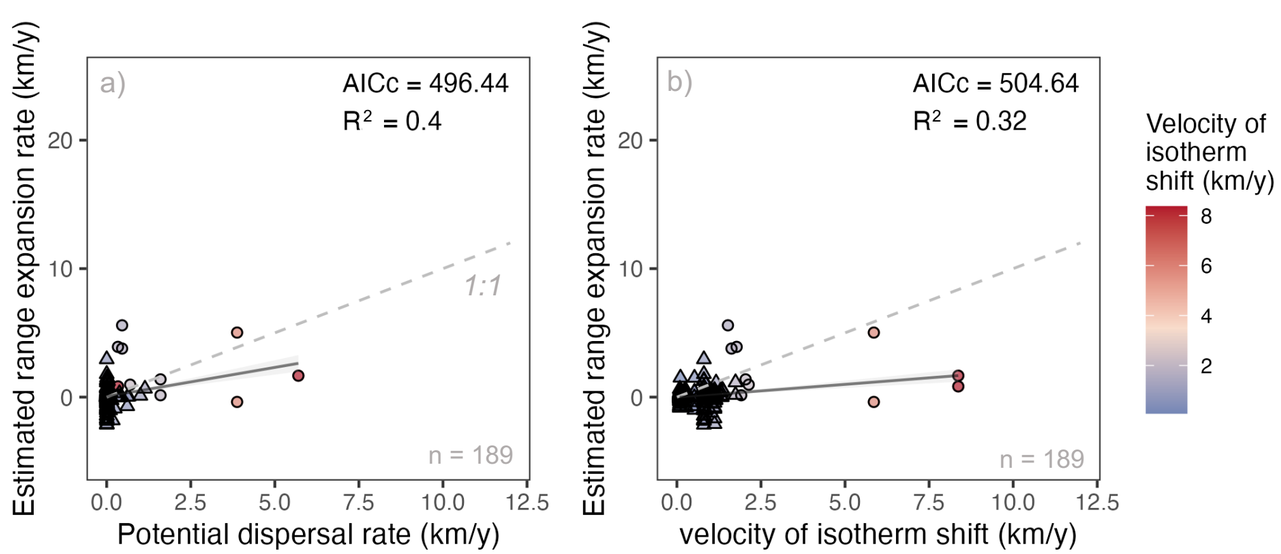

(a)
1:1
Range expansion rate (km/yr)
Potential dispersal rate (km/yr)
(b)
Minimum of potential dispersal rate and velocity of climate change (km/yr)
Velocity of climate change (km/yr)
Dispersal rate < velocity of climate change
Dispersal rate > velocity of climate change

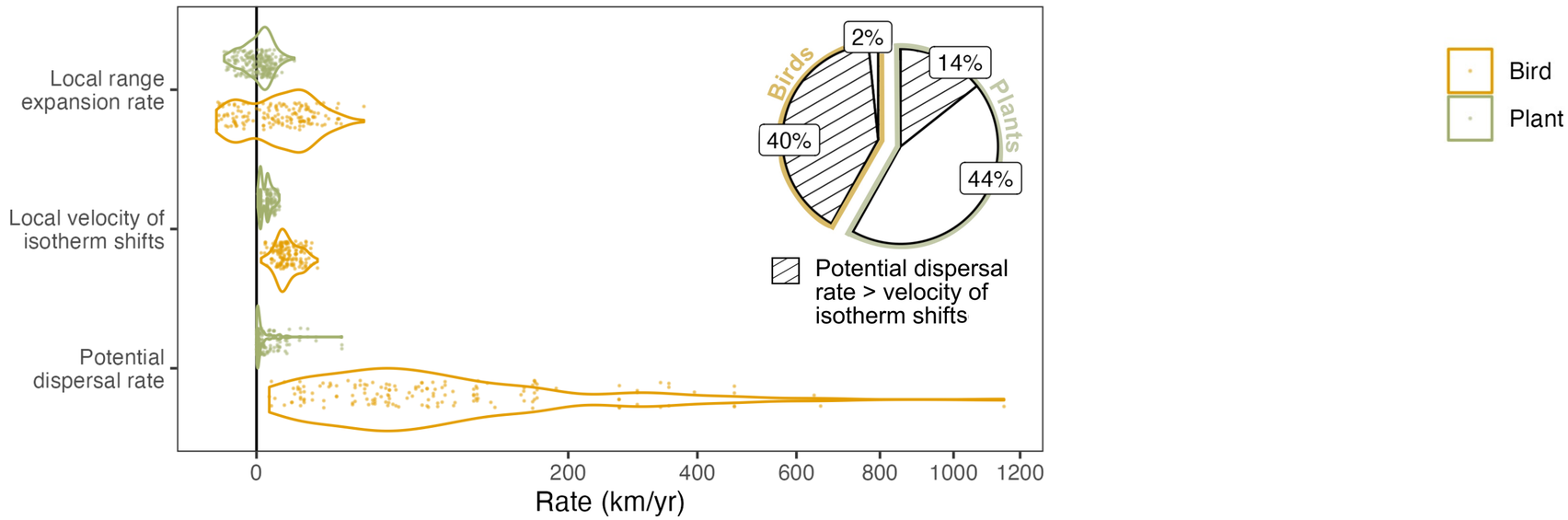

a)
Bird
Mammal
Plant
Squamate
AICc = 797.97
R^2 = 0.1
1:1
n = 211
Observed range expansion rate (km/y)
Potential dispersal rate (km/y)
b)
AICc = 800.02
R^2 = 0.09
n = 211
Velocity of climate change (km/y)
Velocity of climate change (km/y)

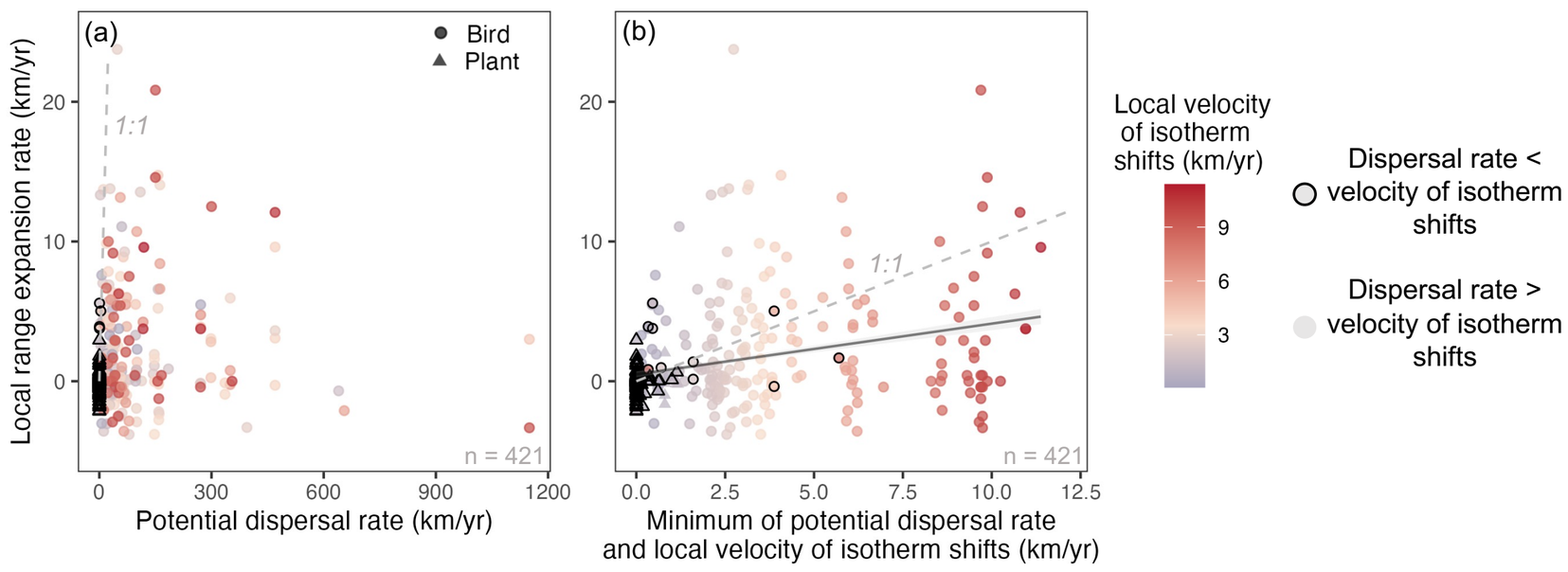

Estimated range expansion rate (km/y)
Potential dispersal rate (km/y)
1:1
n = 421
Birds
Plants
2%
48%
14%
36%
Potential dispersal rate > velocity of isotherm shift
Bird
Plant
Velocity of isotherm shift (km/y)

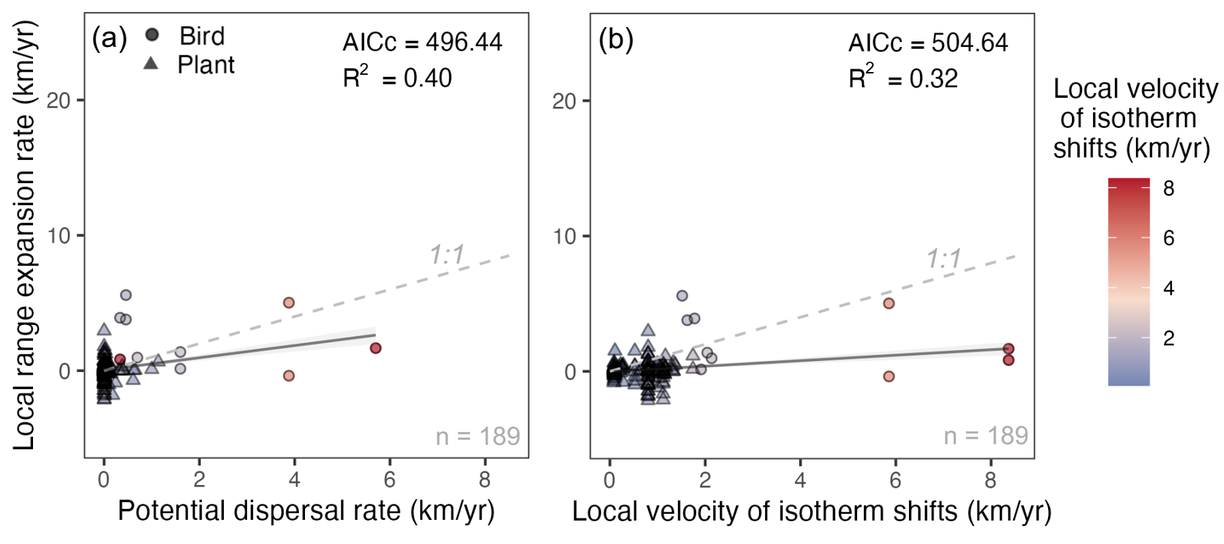

Estimated range expansion rate (km/y)
Minimum of potential dispersal rate and velocity of isotherm shift (km/y)
Bird
Plant
1:1
n = 421
Dispersal rate < velocity of isotherm shift
Dispersal rate > velocity of isotherm shift
Velocity of isotherm shift (km/y)


a)
b)
Local range expansion rate (km/yr)
Potential dispersal rate (km/yr)
Minimum of potential dispersal rate and velocity of climate change (km/yr)
1:1
Velocity of climate change (km/yr)
Dispersal rate < velocity of climate change
Dispersal rate > velocity of climate change


a)
AICc = 496.44
R² = 0.4
n = 189
Estimated range expansion rate (km/y)
Potential dispersal rate (km/y)
1:1
b)
AICc = 504.64
R² = 0.32
n = 189
velocity of isotherm shift (km/y)
Velocity of isotherm shift (km/y)


Local range expansion rate
Local velocity of isotherm shifts
Potential dispersal rate
0
200
400
600
800
1000
1200
Rate (km/yr)
Birds
Plants
2%
14%
40%
44%
Potential dispersal rate > velocity of isotherm shifts
Bird
Plant


(a)
Bird
Plant
1:1
n = 421
Local range expansion rate (km/yr)
Potential dispersal rate (km/yr)
(b)
1:1
n = 421
Minimum of potential dispersal rate and local velocity of isotherm shifts (km/yr)
Local velocity of isotherm shifts (km/yr)
Dispersal rate < velocity of isotherm shifts
Dispersal rate > velocity of isotherm shifts


(a)
Bird
Plant
AICc = 496.44
R^2 = 0.40
1:1
n = 189
Local range expansion rate (km/yr)
Potential dispersal rate (km/yr)
(b)
AICc = 504.64
R^2 = 0.32
1:1
n = 189
Local velocity of isotherm shifts (km/yr)
Local velocity of isotherm shifts (km/yr)